EXAM PAPERS PRACTICE

**Q1.**

(a) Obtain the general solution of the differential equation

$$\tan x \frac{dy}{dx} + y = \sin x \tan x$$

where $0 < x < \frac{\pi}{2}$

**(5)**

(b) Hence find the particular solution of this differential equation, given that $y = \frac{1}{2\sqrt{2}}$ when $x = \frac{\pi}{4}$

**(2)**
**(Total 7 marks)**

**Q2.**

(a) Show that $\tan x$ is an integrating factor for the differential equation

$$\frac{dy}{dx} + \frac{\sec^2 x}{\tan x}y = \tan x$$

**(2)**

(b) Hence solve this differential equation, given that $y = 3$ when $x = \frac{\pi}{4}$.

**(6)**
**(Total 8 marks)**



**Q3.**

(a) Differentiate $\ln(\ln x)$ with respect to $x$.

**(1)**

(b) (i) Show that $\ln x$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} + \frac{1}{x \ln x} y = 9x^2, \quad x > 1$$

**(2)**

(ii) Hence find the solution of this differential equation, given that $y = 4e^3$ when $x = e$.

**(6)**
**(Total 9 marks)**

**Q4.**

(a) By using an integrating factor, find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{2}{x}y = \ln x$$

**(7)**

(b) Hence, given that $y \to 0$ as $x \to 0$, find the value of $y$ when $x = 1$.

**(3)**

**(Total 10 marks)**

## Q5.

(a) By using an integrating factor, find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{4}{2x+1}y = 4(2x+1)^5$$

giving your answer in the form $y = f(x)$.

**(7)**

(b) The gradient of a curve at any point $(x, y)$ on the curve is given by the differential equation

$$\frac{dy}{dx} = 4(2x+1)^5 - \frac{4}{2x+1}y$$

The point whose $x$-coordinate is zero is a stationary point of the curve. Using your answer to part (a), find the equation of the curve.

**(3)**

**(Total 10 marks)**



## Q6.

By using an integrating factor, find the solution of the differential equation



$$\frac{dy}{dx} - \frac{2}{x}y = 2x^3e^{2x}$$

given that $y = e^4$ when $x = 2$. Give your answer in the form $y = f(x)$.

**(Total 9 marks)**

## Q7.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + (\cot x)y = \sin 2x, \qquad 0 < x < \frac{\pi}{2}$$

given that $y = \frac{1}{2}$ when $x = \frac{\pi}{6}$

**(Total 10 marks)**

**Q8.**

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + \frac{3}{x}y = (x^4 + 3)^{\frac{3}{2}}$$

given that $y = \frac{1}{5}$ when $x = 1$.

**(Total 9 marks)**

**Q9.**

(a) Show that $\frac{1}{x^2}$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} - \frac{2}{x}y = x$$

**(3)**

(b) Hence find the general solution of this differential equation, giving your answer in the form $y = f(x)$.

**(4)**

**(Total 7 marks)**

**Q10.**

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} - y\tan x = 2\sin x$$



given that $y = 2$ when $x = 0$.



**(Total 9 marks)**

**Q11.**

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + \frac{4x}{x^2 + 1}y = x$$

given that $y = 1$ when $x = 0$. Give your answer in the form $y = f(x)$.

**(Total 9 marks)**

**Q12.**

(a) A differential equation is given by

$$x\frac{d^2y}{dx^2} - \frac{dy}{dx} = 3x^2$$

Show that the substitution

$$u = \frac{dy}{dx}$$

transforms this differential equation into

$$\frac{du}{dx} - \frac{1}{x}u = 3x$$

**(2)**

(b) By using an integrating factor, find the general solution of

$$\frac{du}{dx} - \frac{1}{x}u = 3x$$

giving your answer in the form $u = f(x)$.

**(6)**

(c) Hence find the general solution of the differential equation

$$x\frac{d^2y}{dx^2} - \frac{dy}{dx} = 3x^2$$

giving your answer in the form $y = g(x)$.

**(2)**
**(Total 10 marks)**

## Q13.

(a) Show that $x^2$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} + \frac{2}{x}y = 3(x^3+1)^{\frac{1}{2}}$$



**(3)**

(b) Solve this differential equation, given that $y = 1$ when $x = 2$.

**(6)**
**(Total 9 marks)**

## Q14.

By using an integrating factor, find the solution of the differential equation

$$\frac{dy}{dx} + (\tan x)y = \sec x$$

given that $y = 3$ when $x = 0$.

**(Total 8 marks)**

## Q15.

(a) The function $y\,(x)$ satisfies the differential equation

$$\frac{dy}{dx} = f\,(x, y)$$

where

$$f\,(x, y) = x \ln x + \frac{y}{x}$$

and

$y\,(1) = 1$

(i) Use the Euler formula

$$y_{r+1} = y_r + h\,f\,(x_r, y_r)$$

with $h = 0.1$, to obtain an approximation to $y\,(1.1)$.

**(3)**

(ii) Use the formula

$$y_{r+1} = y_{r-1} + 2h\,f\,(x_r, y_r)$$

with your answer to part (a)(i) to obtain an approximation to $y\,(1.2)$, giving your answer to three decimal places.

**(4)**

(b) (i) Show that $\frac{1}{x}$ is an integrating factor for the first-order differential equation

$$\frac{dy}{dx} - \frac{1}{x}y = x \ln x$$



**(3)**

(ii) Solve this differential equation, given that $y = 1$ when $x = 1$.

**(6)**

(iii) Calculate the value of $y$ when $x = 1.2$, giving your answer to three decimal places.

**(1)**

**(Total 17 marks)**

## Q16.

(a) Show that $\sin x$ is an integrating factor for the differential equation

$$\frac{dy}{dx} + (\cot x)y = 2 \cos x$$

**(3)**

(b) Solve this differential equation, given that $y = 2$ when $x = \frac{\pi}{2}$.

**(6)**
**(Total 9 marks)**

## Q17.

(a) Show that the substitution

$$u = \frac{dy}{dx} + 2y$$

transforms the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^{-2x}$$

into

$$\frac{du}{dx} + 2u = e^{-2x}$$

**(4)**

(b) By using an integrating factor, or otherwise, find the general solution of

$$\frac{du}{dx} + 2u = e^{-2x}$$

giving your answer in the form $u = f(x)$.

**(5)**

(c) Hence find the general solution of the differential equation



$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^{-2x}$$

giving your answer in the form $y = g(x)$.

**(5)**
**(Total 14 marks)**